MCQ 1: Understand and Choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. $(x^3)\prime$ value
   - A. x^2
   - B. $3x^2$
   - C. $3x$
   - D. $\frac{x^4}{4}$
2. $(\frac{1}{x})\prime$ value
   - A. $-\frac{1}{x^2}$
   - B. $\frac{1}{x^2}$
   - C. $-\frac{1}{x}$
   - D. 1
3. $(uv)\prime$ value
   - A. $u\prime v\prime$
   - B. $u\prime + v\prime$
   - C. $uv\prime$
   - D. $u\prime v + uv\prime$
4. $f\prime > 0$ over an interval implies
   - A.f. positive
   - B. f. negative
   - C. f. Strictly increasing
   - D. f. constant
5. $\frac{x^2+1}{x-1}$ is defined if
   - A. $x > 0$
   - B. $x \neq 1$
   - C. $x \neq -1$
   - D. $x \neq 0$

MCQ 2: understand and choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. $x^3 - x$ is

   A. Pair

   B. constant

   C. odd

   D. Neither pair nor odd

2. $f\prime(a) = 0$ necessarily implies

   A. Horizontal tangent

   B. Maximum

   C. Minimum

   D. f. constant

3. Derivative from $\sqrt{x}$:

   A. $2\sqrt{x}$

   B. $\dfrac{1}{\sqrt{x}}$

   C. $\dfrac{\sqrt{x}}{2}$

   D. $\dfrac{1}{2\sqrt{x}}, x > 0$

4. Minimum of $x + \dfrac{1}{x}$ on]0; + ∞[:

   A. 0

   B. 2

   C. 1

   D. None

5. Newton for $x^2 - 2$ is not defined with $x_0 =$

   A. 1

   B. 2

   C. 0

   D. -1

True or false?

1. The derivative of a positive function is positive.

2. An inner extremum of a drifting function gives a zero derivative.

3. The output of two increasing functions is increasing.

Personal review

I drift on the right domain; I simplify without losing the forbidden values; I distinguish extremum
local and global; I sometimes choose factorization rather than derivative.
For each objective, indicate: acquired without help; succeeded with help; to rework. Write an example and
a mistake to avoid.